

The Mathematician's Secret

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Category: Cryptography

Difficulty: Medium-Hard

Introduction

Welcome, young apprentice! You have stumbled upon an ancient **spellbook** filled with **mysterious numbers**.

Legends say that the **Mathematician** himself encoded a **hidden message** within these numbers using the magic of **linear equations** and **modular arithmetic**.

Your task is to **reverse the spell** and uncover the true message.

The Challenge

Inside **spellbook.txt**, you will find a **list of encrypted numbers**.

The Mathematician used a **linear transformation** to disguise the original text:

$$\begin{aligned} ax + by &\equiv c1 \pmod{2^{64}} \\ cx + dy &\equiv c2 \pmod{2^{64}} \end{aligned}$$

But beware! The values of a, b, c, d are **unknown**.

Can you **discover the hidden equations** and solve for x and y ?

Provided Files

spellbook.txt → Contains the **ciphertext**.

Hints

1. Linear equations can be solved if you have enough information.

Goal

Your objective is to **decrypt the hidden message** and recover the flag.

Submit your answer in this format: `Alphabit{mathemagical_solution}`

Good luck.